

ILDAR SHAYAKHMETOV

Senior Backend & AI-Systems Engineer | High-Load Fintech

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SUMMARY

Senior Backend & AI-Systems Engineer with 6+ years designing high-load, distributed backend systems. Expert in Java, Kotlin, and Scala with reactive programming (Project Reactor, Akka Streams) and deep fintech experience — payments, money movement, and PCI-DSS-compliant services. Also architects event-driven AI platforms in Rust and Python (algorithmic trading, AI health coaching). Track record of leading projects, optimizing performance, and keeping systems stable under load.

KEY ACHIEVEMENTS

- Engineered a real-time streaming platform processing **5M+ messages/sec** from Kafka (Java + Akka Streams).
- Cut cloud spend by **~EUR 20k/month** via a GraalVM native-image migration.
- Delivered **PCI-DSS-compliant** payment services in production (Kotlin, Project Reactor).
- Architected an event-driven algorithmic-trading platform with an **architecturally unbypassable risk engine** (Rust + Python + Kafka).

EXPERIENCE

Software Developer — Paysend

Feb 2023 – Present

Belgrade, Serbia

- Develop and maintain high-load payment services in Kotlin and Java; reactive architecture with Project Reactor.
- Ensure PCI-DSS compliance and transaction security; integrate external payment systems.
- Manage inter-service messaging over RabbitMQ; drove a GraalVM native-image migration saving ~EUR 20k/month.
- Build CI/CD pipelines (GitLab) with automated testing (JUnit, TestContainers).
- Take part in technical hiring — conducting interviews — and in prioritizing the team's backlog.

Software Engineer — Perfect Art

May 2022 – Feb 2023

Belgrade, Serbia

- Architected a financial-activity screening system for small businesses.
- Implemented core logic in Scala + Akka for a highly concurrent, resilient solution.
- Coached teammates on the Scala/Akka codebase and the screening domain to raise delivery quality.

Software Engineer — MTS

May 2020 – May 2022

Moscow, Russia

- Designed a real-time, high-load streaming platform processing **5M+ messages/sec** from Kafka.
- Built the core processing engine in Java + Akka Streams; wrote consumer services in Scala (ScalaTest).
- Used Aerospike for caching and deduplication; managed inter-service communication through Kafka.
- Mentored interns and new hires; delivered internal Scala lectures to accelerate team ramp-up.

Software Engineer — Raiffeisen Bank

May 2018 – Aug 2019

Moscow, Russia

- Built and enhanced a high-load notification service in Java + Spring Integration; rewrote the mass-mailing flow.
- Redesigned caching mechanisms to improve performance; built a notification-history service from scratch.
- Contributed to frontend work with React and WebSockets for real-time communication.

Java & Android roles at Alfa-Bank, EPAM Systems, Trendsoft, and Dixy — core banking (OpenID Connect authentication), high-load data-validation systems and REST APIs, system integrations (1C, AmoCRM), and mobile development (Retrofit, RxJava).

SKILLS

- **Languages:** Kotlin, Java, Scala, Rust, Python, TypeScript, SQL
- **Frameworks & Reactive:** Spring Boot, Spring Cloud Gateway, Spring WebFlux, Project Reactor, Akka Streams, ZIO, ZIO HTTP, Tokio, RSocket, NestJS
- **Messaging & Streaming:** Apache Kafka, RabbitMQ, RSocket
- **Data & Caching:** PostgreSQL, MongoDB, Aerospike, Redis, Quill
- **AI & Event-Driven:** event-driven architecture, gRPC, FlatBuffers, LLM-backed systems, auditable/reproducible decision pipelines
- **DevOps & Tooling:** Docker, Kubernetes, GitLab CI/CD, Gradle, GraalVM, TestContainers, JUnit, Git

PROJECTS

- **Alladin** — Event-driven algorithmic-trading platform with an architecturally unbypassable risk engine and reproducible, auditable decisions. *Rust, Tokio, Kafka, gRPC.*
- **HealthTracer** — AI health coach built as a system, not a chatbot: a Postgres-backed structured domain model is authoritative while the AI emits *typed proposals* (never direct DB writes) that the backend validates and the user approves, each an immutable revision. Core is a multi-domain fan-out & synthesis LLM pipeline behind a deterministic planner that clamps output to a hard capability allowlist with safety floors enforced in code. *TypeScript, NestJS, Next.js, Expo, Drizzle/PostgreSQL.*
- **bot-constructor** — Microservices automation platform for visually building bots with an n8n-style workflow engine: a React Flow drag-and-drop node-graph UI over a reactive Kotlin backend split into four services (gateway, auth, client-api, bot-api). An item-based execution engine runs 23 node types — triggers, branches, HTTP, sandboxed code, and live connectors to Claude, Telegram, Slack, and Discord — with a GraalVM JavaScript expression/code evaluator, an AES-256-GCM-encrypted credential store, per-node execution history, and cron scheduling. RSocket binary transport, Spring WebFlux non-blocking throughout, reactive MongoDB, JWT auth; Docker Compose + Kubernetes. *Kotlin, Spring Boot 4, Spring Cloud Gateway, RSocket, WebFlux, MongoDB, GraalVM, React 19 / React Flow.*
- **zio-avito-desk** — Full-stack classifieds board built functional-first: a layered Scala 3 + ZIO backend (domain/storage/service/route/server) with ZIO HTTP, compile-time type-safe SQL via Quill, and ZLayer dependency injection. Session-cookie auth with bcrypt-hashed passwords, ownership-gated CRUD, image uploads, PostgreSQL with Flyway-versioned migrations, and 33 Testcontainers-backed tests; React 19 / TypeScript frontend with filter/search and GitHub Actions CI. *Scala 3.3, ZIO 2, ZIO HTTP 3, Quill, PostgreSQL, Flyway, React.*
- **BeoSand** — Telegram-first platform for booking volleyball courts in Belgrade: a React admin console (scheduling, court-load heatmap, booking requests, broadcasts) plus a Telegram mini-app for players. Server-authoritative by design — pricing, availability, and capacity are computed server-side; Telegram Login auth, RSD pricing, RU/SR/EN i18n. *TypeScript, React + Vite, Telegram Mini Apps.*

EDUCATION

HSE University — B.Sc., Software Engineering · 2014 – 2018

CERTIFICATIONS

- JVA-074 Java Advanced: Functional, Asynchronous and Reactive Programming
- Kotlin Developer

LANGUAGES

- English — Advanced (C1)
- Russian — Native